

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.4 CHEMICAL BONDING: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Think back to the Carbon Paradox from Lesson 1: a diamond chip from a Nairobi jeweller's cutting tool and a pencil graphite stick, both labelled '100% Carbon'. Graphite smeared a grey mark on the ceramic tile and lit the bulb, while diamond scratched the tile and did not conduct. Answer each section in the blank space under the prompt. Write in full sentences, use words such as valence electrons, covalent bond, delocalised electron, Van der Waals forces, ionic, metallic, lattice and structure, and support every claim with evidence from the lessons (use Claim, Evidence, Reasoning where you can). You may draw labelled diagrams. Where a section asks you to use the table, read it carefully first.

Section 1: Valence Electrons and Bond Type

Carbon has 6 electrons in total, but only some of them take part in bonding.

(a) Which electrons take part in bonding, and how many does carbon have? How would you show them in a Lewis dot diagram?

(b) Explain how the octet rule helps you predict the number of bonds carbon forms.

(c) Atoms can gain, lose or share electrons. Explain how valence electrons decide whether a bond is ionic, covalent or metallic, using NaCl and a covalent molecule such as CH₄ as examples.

Write your answer here:

Section 2: Explaining the Carbon Paradox

Diamond and graphite are both made only of carbon atoms.

(a) Describe how carbon atoms are bonded in diamond and how they are bonded in graphite. Include the number of bonds per carbon atom and the shape of the arrangement.

(b) Use your answer to (a) to explain why graphite conducts electricity but diamond does not.

(c) Explain why diamond scratched the ceramic tile while graphite smeared on it. Your answer must mention the forces inside the structure.

Write your answer here:

Section 3: Intermolecular Forces and Metallic Bonding

- (a) Water boils at 100 °C while H₂S boils at -60 °C, even though H₂S is a larger molecule. What does this show about the forces between water molecules, and what is the cause?
- (b) Iodine changes directly from solid to purple vapour when warmed gently. Name the forces holding iodine molecules together and explain why a non-polar substance still has these forces.
- (c) Describe the bonding in aluminium foil and use it to explain why aluminium conducts electricity and can be bent without breaking.

Write your answer here:

Section 4: Bond Type, Structure and Properties

Use the table of results from your investigation to answer the questions below.

Substance	Structure type	Dissolves in water	Conducts as solid	Conducts when dissolved
NaCl	Giant ionic	Yes	No	Yes
Sugar	Simple molecular	Yes	No	No
SiO ₂ (sand)	Giant covalent	No	No	Not applicable
Aluminium	Giant metallic	No	Yes	Not applicable
Graphite	Giant covalent	No	Yes	Not applicable

- (a) Explain why NaCl conducts only when dissolved, but not as a solid.
(b) Why does sugar not conduct, even when dissolved?
(c) Graphite and SiO₂ are both giant covalent, yet only graphite conducts. Explain this.
(d) Which type of structure would you expect to have the lowest melting points, and why?

Write your answer here:

Section 5: From Structure to Use

Answer the driving question: Why do diamond and graphite, both made only of carbon atoms, have completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for?

(a) Write a Claim, Evidence and Reasoning paragraph that answers the driving question using diamond and graphite.

(b) Give one use of diamond and one use of graphite in Kenya and link each use to its structure.

(c) A student suggests that a copper wire or an NaCl solution could replace graphite in a dry-cell battery. Choose one and evaluate this idea using bonding ideas.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of valence electrons and bond types	Accurately explains how valence electrons and the octet rule lead to ionic, covalent and metallic bonding, with correct examples such as NaCl and CH ₄ .	Identifies valence electrons and the three bond types with mostly correct examples but limited explanation of why each forms.	Confuses bond types or valence electrons, or gives examples without explanation.
Explanation of diamond and graphite	Clearly links number of bonds, geometry, delocalised electrons and Van der Waals forces to conductivity and hardness of both substances.	Explains most differences between diamond and graphite but misses one link, such as the weak forces between layers.	States that the properties differ without linking them to the bonding or structure.

Use of evidence and data	Uses the table and lesson observations accurately to support every claim, including the graphite anomaly.	Uses some evidence correctly but some claims are unsupported.	Gives little or incorrect evidence.
Structure-property-use reasoning	Connects bond type to structure, properties and a real Kenyan use, and applies the ideas to a new situation.	Connects structure to properties and uses, with limited application to a new situation.	Lists uses with no reasoning from structure.
Scientific language and communication	Uses correct terms precisely and organises answers in clear Claim, Evidence, Reasoning form.	Uses mostly correct terms with generally clear organisation.	Uses few scientific terms or the answers are unclear.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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